USE CASE 01

FAQ Agent (also RAG)

Automating Customer Questions for E-Commerce

Industry: E-Commerce · Product: Laptops & Mobiles

How an AI Agent Works

The core components that power every AI agent

Chat Interface

The entry point. Customers send and receive messages through this layer.

Brain (LLM)

The Large Language Model at the core. Understands input and generates responses.

Chat History

Gives the agent memory — it knows exactly what was said earlier in the conversation.

Tools

External capabilities the agent can invoke to perform specific actions.

System Prompt

The agent's instruction set. Defines its role, tone, and boundaries.

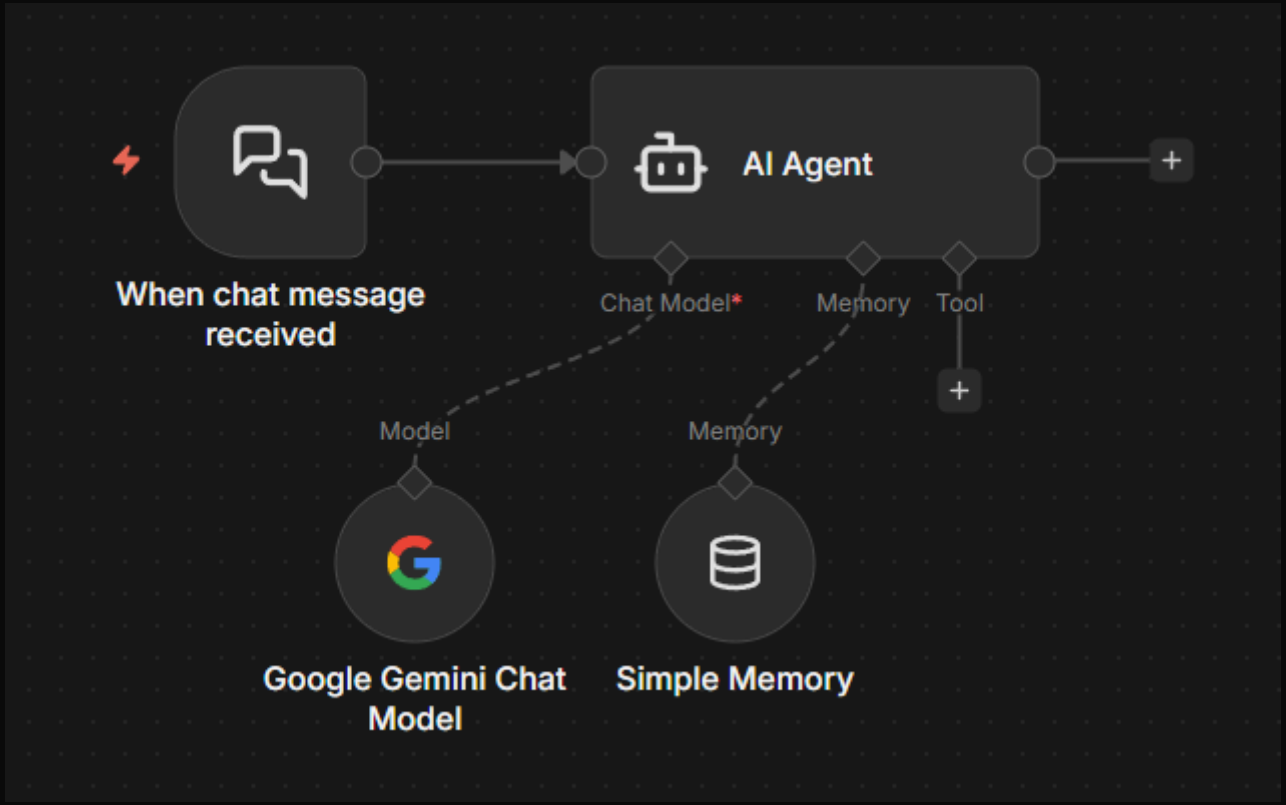
Knowledge Base

TODAY'S FOCUS

Additional working memory. Stores company-specific information the agent needs to answer accurately.

Basic AI Agent

We will be going fast on

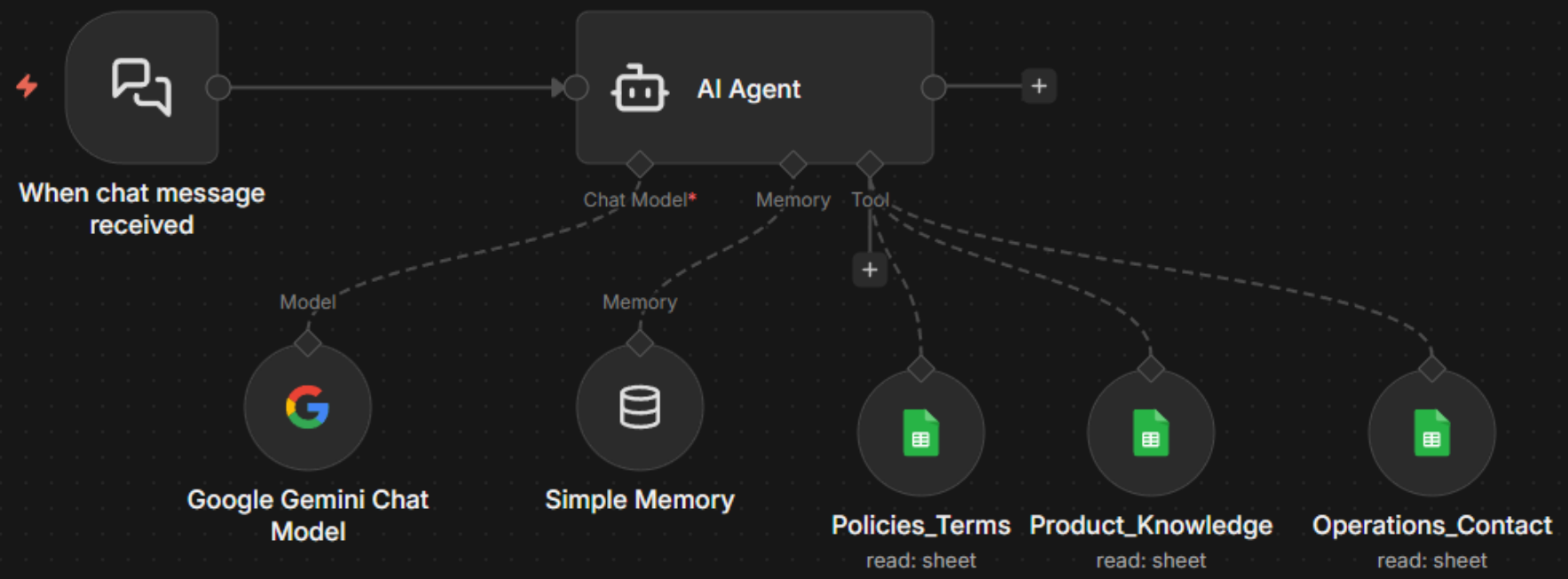


FAQ Agent (RAG)

With Knowledge Base (Can Easily be replaced with

Questions to ask

1. What is the return policy?
2. Do you have earbuds?
3. How to get in touch?



“

It is this additional **knowledge base** that we are going to be working on today.

The Situation

Imagine you run an e-commerce company selling laptops & mobiles.

Profit = Revenue − Costs + Customer Experience

Revenue

How you earn

- Product range & pricing
- Conversion rate
- Repeat purchases & upsells

Costs

What you spend

- Inventory & logistics
- Returns & refunds
- Support operations

Customer Experience

How customers interact with you

- Pre-sale product questions
- Order tracking & updates
- After-sales support & complaints

Where the FAQ Agent Creates Value

The FAQ Agent touches all three business buckets — simultaneously.

Revenue

How you earn

- Product range & pricing
- Conversion rate

+20–25%

conversion uplift

Spec question at 11pm → instant answer → £1,200 laptop purchased within minutes

Costs

What you spend

- Support operations
- Returns & refunds

£1.2M/yr

support costs saved

500 tickets/day × £10 avg · Agent handles 65% → £3,250/day saved

Customer Experience

How customers interact

- Pre-sale questions
- After-sales support

+20–30%

CSAT improvement

Midnight query on return policy → instant answer → purchase completed, not abandoned

Profitability

Revenue uplift + Cost reduction hit profitability simultaneously, from both sides

£1.5M

direct profitability impact

On £10M/yr turnover: 3% conversion lift = £300K extra revenue + £1.2M cost savings — without adding headcount

Five Questions. Five Support Areas.

Website chat widget · WhatsApp number on site — same agent, same knowledge base, two channels.

"I ordered a Samsung phone 3 days ago. Where is it?"	→	**1 Order Tracking**
"Does the MacBook Air M3 support an external 4K monitor?"	→	**2 Product Features**
"Is there a student discount on any of your laptops?"	→	**3 Pricing & Discounts**
"My laptop freezes every time I open Chrome. What should I do?"	→	**4 Technical Support**
"I received the wrong item. Can I return it and get the right one?"	→	**5 Return Policy**

Understanding Intent

Customer Message → **Agent Reads** → **Intent Identified** → **KB Searched** → **Answer**

Customer Says	Keywords Found?	True Intent
"I want to send this back"	None — no "return" or "refund"	**Return Policy**
"This device is acting up"	None — no technical terms	**Technical Support**
"I'm not happy with this purchase"	None — no keywords at all	**Return Policy**

The Keyword Problem

- Looks for exact keywords: "return", "refund", "policy"
- No match → generic response or wrong bucket
- Customer gets the wrong answer — or no answer
- Trust is lost. Sale is gone.

What the LLM Does Instead

- Reads the full sentence in context
- Understands meaning — not just matching words
- Classifies intent even when keywords are absent
- This is the Brain (Slide 02) doing its real work

The System Prompt as a Router

System Prompt

You are an FAQ agent for TechMart,

an e-commerce store selling laptops and mobiles.

Answer customer questions using

→ Returns, refunds, exchanges → use: **Return Policy KB**

only the knowledge base provided.

→ Product specs, compatibility → use: **Product Features KB**

When a customer speaks about:

→ Pricing, offers, discounts → use: **Pricing & Discounts KB**

→ Order status, shipping, delivery → use: **Order Tracking KB**

→ Device errors, faults, setup → use: **Technical Support KB**

Always be helpful. Do not guess.

Knowledge Base

Return Policy

Return window, conditions, exceptions, how to initiate

Product Features

Specs, compatibility, comparisons, what each product supports

Pricing & Discounts

Current offers, student/corporate deals, promo codes

Order Tracking

Timelines, carrier info, how to track, delays

Technical Support

Troubleshooting, common errors, warranty, escalation path

System Prompt = Router · Knowledge Base = Library · LLM = Reader

How It All Comes Together

Customer: *"I got a laptop last week and the keyboard keeps skipping letters. Can I swap it for a new one?"*

STEP 1

Intent Detected

"last week"

Return window → relevant

"keyboard keeps skipping"

Hardware defect → Tech Support

"Can I swap it"

Exchange → Return Policy

Dual intent — Tech Support + Return Policy

STEP 2

System Prompt Routes

Technical Support KB

Is this a known keyboard issue with a fix?

Return Policy KB

Is the customer within the 15-day swap window?

STEP 3

Agent Responds

"Sorry to hear about the keyboard! This can sometimes be a driver issue — here are two quick steps to try. If the problem continues, you're within our 15-day swap window. Here's how to request a replacement: [link]"

One message · Two intents · Two KBs · One right answer · No human. No delay.

A Note for Our

Visionaries

If you are finding it difficult to follow along, it is likely because you have not yet completed the three entry-point sessions. These are not optional — they are the foundation everything else builds on.

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01 AI Agent Accelerator — Onboarding Video

02 AI Agent Kickstarter

03 Basic Agent Session

We will be moving fast — to respect everyone's time. If you have not completed these today, please do so before the next session.

Preparation is how you show up.

Building the Agent — Step by Step

Four nodes. That is the entire agent skeleton.

Chat Node

Receives the customer's message — the user's entry point

→

Agent Node

The orchestrator — manages the workflow and all components

→

Brain (LLM)

The reasoning engine — Claude or GPT connected to Agent Node

Simple Memory

Remembers prior messages in the session

These four nodes map directly to the framework in Slide 02 — we are now building what we described.

1	**Chat Node**	Receives the message
2	**Agent Node**	Orchestrates all components
3	**Brain (LLM)**	Reads intent and generates the response
4	**Simple Memory**	Remembers the conversation so far

Connecting the Knowledge Base

The knowledge base is Google Sheets. Start with one, confirm it works, then copy-paste.

Step 1 — First Sheet + First Knowledge Area

Add a Google Sheets connection to the Agent Node. Sheet 1 = Q&A pairs for your first support topic. This becomes Knowledge Area 1.

Step 2 — Second Sheet + Second Knowledge Area

Copy-paste the first connection. Sheet 2 = Q&A pairs for your second support topic. Repeat this pattern for all five categories.

Step 3 — Add the System Prompt

Paste the System Prompt directly into the Agent Node. It tells the agent which sheet to use for which intent — the routing logic from Slide 08, now live.

Full Agent Structure

```
Agent Node
├── Brain (LLM)
├── Simple Memory
├── Google Sheet 1  →  Return Policy KB
├── Google Sheet 2  →  Product Features KB
├── Google Sheet 3  →  Pricing & Discounts KB
├── Google Sheet 4  →  Order Tracking KB
├── Google Sheet 5  →  Technical Support KB
└── System Prompt   →  "If intent is X, use Sheet Y"
```

Your Starting Point — and How to Own It

What's on GitHub

Pre-built System Prompt

Ready to paste directly into your Agent Node

Google Sheet Template

Five tabs with sample Q&A already filled in — just copy it

Three Steps to Make It Yours

1 **Change the System Prompt**

Update the agent's identity, routing rules, and tone to match your brand

2 **Update the Q&A in the Sheets**

Replace the sample answers with your actual business content

3 **Rename the Sheets**

Give them names that match your business — the agent is now entirely yours

Follow along as-is today. Customise at your own pace after.

How to Deploy Your FAQ Agent

One agent. Three deployment options. *Same knowledge base. Same System Prompt.* Different front door.

TODAY

S

Standalone (n8n)

Deploy directly via n8n's built-in chat interface

Demonstrated live today

T

Telegram

Connect the agent to a Telegram bot using the bot API

AI Agent Kickstarter course

W

WhatsApp

Connect the agent to a WhatsApp number shared on your site

Dedicated WhatsApp integration video